

Genomic Variant Interpretation Platform

Team Handover

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Purpose: A practical introduction for any Team Nucleo member who needs to explain, run, demo, or extend the project.

***Important:** this is an educational/prototype decision-support application. The model is trained on synthetic data and its output must never be used as a standalone clinical diagnosis.*

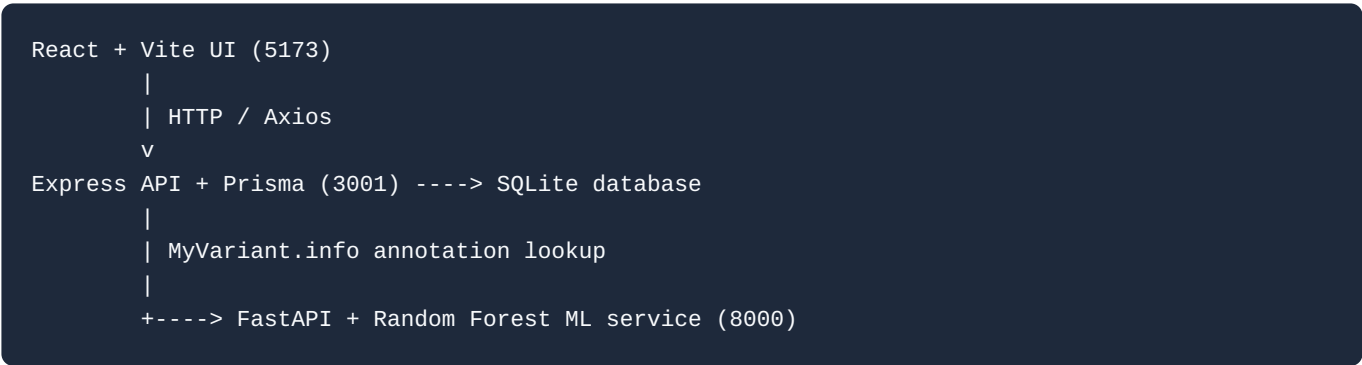
1. What the project does

The platform accepts a Variant Call Format (VCF) file, stores its variants, enriches selected variants with public genomic evidence, estimates pathogenicity with a machine-learning service, and presents the findings in an interactive review dashboard and downloadable PDF.

The user flow is:

1. Upload a `.vcf` file.
2. The backend parses it line by line and stores the patient/case and variants in SQLite.
3. Select **Run Variant Analysis**.
4. The backend retrieves allele frequency, CADD, and ClinVar information from MyVariant.info (with safe fallbacks), then sends numerical features to the Python ML service.
5. The ML service returns a probability, classification, and explanation. The backend persists these as evidence and updates the variant status.
6. The frontend lets the reviewer search/filter variants, inspect an evidence drawer, and export priority findings to a PDF report.

2. Architecture



Why this architecture?

- **React frontend:** fast, component-based interface for a table-heavy review workflow.
- **Express API:** handles uploads, validation, database access, and service orchestration in JavaScript.
- **FastAPI ML microservice:** keeps Python/scikit-learn code separate from Node.js while exposing a simple HTTP contract.
- **REST APIs:** make frontend/backend/ML components independently testable and replaceable. They also model how real clinical systems integrate distinct services.
- **Prisma + SQLite:** Prisma gives typed, readable database access; SQLite keeps local demos simple with no external database server.

3. Technology stack

Area	Technologies	Role
Frontend	React 19, Vite, Tailwind CSS 4	Dashboard, responsive UI, theme support
Frontend utilities	Axios, Lucide React, jsPDF, jspdf-autotable	API calls, icons, PDF export
Backend	Node.js, Express, Multer, CORS, dotenv	REST API, VCF upload, middleware/configuration
Data layer	Prisma ORM, SQLite	Patient, variant, and evidence persistence
Annotation source	MyVariant.info API	gnomAD allele frequency, CADD score, ClinVar data where available
ML service	Python, FastAPI, Uvicorn, pandas, NumPy, scikit-learn	Prediction endpoint and model serving

Area	Technologies	Role
ML model	RandomForestClassifier	Prototype pathogenicity probability from two features

`xgboost` appears in `ml-service/requirements.txt`, but the active prediction path uses **RandomForestClassifier**, not XGBoost.

4. Inputs, outputs, and data model

Accepted input

- A standard text VCF file with `.vcf` extension.
- Required data columns used by the parser: `CHROM`, `POS`, `REF`, `ALT`; `QUAL` is optional.
- Header/comment lines beginning with `#` are ignored.
- If a record contains multiple ALT alleles, the current parser retains only the first ALT allele.
- Upload size limit: **300 MB**.

The browser currently advertises `.vcf.gz`, but the Multer backend filter accepts only `.vcf`. Treat compressed VCF support as a known improvement item, not a completed feature.

Persisted entities

Entity	Key fields	Meaning
<code>Patient</code>	<code>id</code> , <code>filename</code> , <code>date</code>	One uploaded case/file
<code>Variant</code>	<code>patientId</code> , <code>chrom</code> , <code>pos</code> , <code>ref</code> , <code>alt</code> , <code>qual</code> , <code>status</code>	One called genomic change
<code>Evidence</code>	<code>variantId</code> , <code>frequency</code> , <code>conservation_score</code> , <code>ml_score</code> , <code>shap_explanation</code>	Annotation and model result for one variant

Relationships: one patient has many variants; one variant has at most one evidence record. Deleting a patient cascades to its variants and evidence.

Outputs

- Variant classification: **Benign** , **VUS** , or **Pathogenic** .
- ML score: probability-like value from **0.0** to **1.0** .
- Short feature-based explanation and optional ClinVar note.
- Interactive evidence panel.
- PDF containing a summary, priority findings (Pathogenic + VUS), and interpretation disclaimer.

5. Core features

Case intake and management

- Drag/drop or browse VCF upload.
- Stream-based parsing with **fs** + **readline** , avoiding reading the entire file into memory at once.
- Bulk database insertion in chunks of 1,000 variants.
- Recent-case selector and clear-all patient history action.

Annotation and analysis

- Normalises chromosome/allele values and creates an HGVS-like query for MyVariant.info.
- Extracts gnomAD allele frequency, CADD Phred score, and ClinVar clinical significance when returned.
- Uses concurrent lookup with a 2.5-second request timeout, so an unavailable public API does not freeze the UI.
- Includes a few known prototype/demo loci with deterministic feature values.
- Limits a single analysis run to the first **50 variants** to keep the prototype responsive.
- Sends feature batches to **POST /predict** rather than calling the ML service once per variant.
- Validates the ML classification contract and falls back safely to **VUS** when an unexpected label is received.

Review interface

- Light/dark themes and a custom clinical-genomics visual system.
- Table sorting that prioritises pathogenic variants, status filters, and search by coordinate, allele, status, or available gene text.
- Evidence drawer with score, population frequency, CADD, ClinVar context, and explanation.
- Custom 404 screen, metadata, and a clear empty state.
- Organised client-side PDF export.

6. Basic biology and genomics vocabulary

Term	Team-friendly meaning
DNA	The molecule carrying genetic instructions; it is made of bases A, C, G, and T.
Gene	A DNA region that contributes to a functional product, usually a protein.
Genome	The complete set of genetic material.
Chromosome	A packaged DNA structure; humans normally have 23 pairs.
Variant	A difference in DNA sequence relative to a reference genome. It is not automatically harmful.
Reference / alternate allele	The expected base(s) at a genomic location / the observed changed base(s). Example: G → A .
VCF	Standard tabular text format for storing called variants and metadata.
Allele frequency (AF)	Proportion of people in a population carrying an allele. Very common variants are less likely to explain rare, highly penetrant disorders.
gnomAD	A population genetics resource often used to assess how common a variant is.
CADD Phred score	A computational score intended to rank predicted deleteriousness. Higher values generally indicate more concerning variants, but it is not diagnostic by itself.
ClinVar	Public archive of variant interpretations and supporting clinical assertions. Assertions can conflict or change.
Pathogenic	Evidence supports disease causation for a stated condition/context.
Benign	Evidence supports a non-disease-causing interpretation.
VUS	Variant of uncertain significance; there is not enough or not sufficiently consistent evidence to call it benign or pathogenic.
SHAP	A method normally used to explain model feature contribution. In this prototype, the returned text is a rule-based, feature-oriented explanation rather than a full SHAP-value computation.

7. Model behaviour

The training script creates **10,000 synthetic records** using two features:

- allele_frequency**, sampled log-uniformly from **1e-6** to **0.5**

- `cadd_score` , sampled from `0` to `40`

A biological-inspired logistic formula creates synthetic labels. A Random Forest with 150 trees and maximum depth 8 is trained, then serialized to `ml-service/model.pkl` .

The serving service converts model probability to labels:

Score	Label
<code>< 0.20</code>	Benign
<code>0.20 - <0.80</code>	VUS
<code>≥ 0.80</code>	Pathogenic

These thresholds are a prototype design choice. In real clinical use, they require validation on labelled clinical data, calibration, and governance.

8. Important API endpoints

Method	Endpoint	Used for
GET	<code>/api/health</code>	Backend/database health check
POST	<code>/api/upload</code>	Multipart VCF upload and parsing
POST	<code>/api/analyze/:patientId</code>	Run annotation + ML pipeline
GET	<code>/api/patients</code>	List uploaded cases with variants/evidence
GET	<code>/api/patients/:id</code>	Load one case
DELETE	<code>/api/patients</code>	Clear all local data
GET	<code>/api/variants</code>	Variant retrieval/filtering
GET	<code>/api/evidence/variant/:variantId</code>	Evidence drawer data
GET	<code>http://127.0.0.1:8000/health</code>	ML service health
POST	<code>http://127.0.0.1:8000/predict</code>	Batch feature prediction

9. How to run locally

1. ML service

```
cd ml-service
pip install -r requirements.txt
python -m uvicorn main:app --port 8000 --reload
```

2. Backend

```
cd backend
npm install
npx prisma db push
npm run dev
```

3. Frontend

```
cd frontend
npm install
npm run dev
```

4. Open the frontend URL shown by Vite (normally <http://localhost:5173>).

Health check URLs:

- <http://localhost:3001/api/health>
- <http://127.0.0.1:8000/health>

10. Key files to know

File	Responsibility
frontend/src/App.jsx	Screen state, upload/analysis actions, layout
frontend/src/components/VariantTable.jsx	Search, filters, sorting, row selection
frontend/src/components/EvidenceDrawer.jsx	Detailed evidence display
frontend/src/components/ExportReportButton.jsx	PDF layout and download
backend/routes/upload.js	Upload validation, parsing, storage
backend/services/vcfParser.js	Line-by-line VCF parser
backend/services/annotationService.js	MyVariant.info lookup and feature extraction
backend/services/pipelineService.js	Annotation-to-ML-to-database orchestration
backend/prisma/schema.prisma	Database schema

File	Responsibility
<code>ml-service/main.py</code>	FastAPI prediction service
<code>ml-service/train_model.py</code>	Synthetic-data model training

11. Known limitations and responsible-use notes

- Synthetic training data means reported evaluation metrics do **not** demonstrate clinical performance.
- Only two ML features are used; real interpretation uses phenotype, inheritance, segregation, transcript consequences, coverage/quality, literature, and expert review.
- MyVariant.info can be incomplete, rate-limited, or unavailable. The current fallback produces zero-valued features, which must not be over-interpreted.
- The pipeline processes only the first 50 variants of a patient in this prototype.
- Multi-allelic VCF records keep only the first ALT value.
- This implementation is not designed for protected health information, authentication, audit trails, or clinical compliance.

12. Recommended next work

1. Add genuine `.vcf.gz` decompression or remove the frontend acceptance claim.
2. Replace synthetic data with consented, curated, labelled data and perform external validation/calibration.
3. Include more biologically relevant features and real explainability values.
4. Process all variants through a queued/background job with progress reporting.
5. Add authentication, access control, encrypted storage, audit logs, and deployment configuration before handling real patient data.
6. Add tests for malformed VCF files, annotation timeouts, ML contract failures, PDF export, and end-to-end analysis.